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Hendrik Ebbers

- Hiero Technical Steering Committee
- LFDT Technical Advisory Committee Vice-chair



ADOPTIUM

Java Champions

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Hendrik Ebbers

- Java Champion
- Eclipse Foundation Board
- Founder of Open Elements
- Contractor for Hashgraph

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Java Champions

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Hendrik Ebbers

- I ❤️ Star Wars
- I ❤️ dogs
- I ❤️ boardgames
- I ❤️ open source



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Java Champions

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About the course

- Separated in 3 main parts
- Mix between lecture and hands on
- We will use Java and learn Solidity as programming languages
- All slides are in Englisch

Part 1

- The concept of crypto currency, blockchains and distributed systems
- Understand the benefits and problems with crypto currencies and blockchains
- Understand enterprise use cases
- Understand the general technology behind it

Part 2

- The concept of distributed ledgers in general
- Learn about Ethereum, Hedera and other ledgers that are based on the EVM
- Understand what a smart contract is
- Learn how to create contracts by using Solidity

—Part 3

- Concentrate on Hedera to learn about enterprise use cases
- Understand the global picture of
- Learn how to create contracts by using Solidity
- Understand how all parts come together
- Create a real world application

The goal of the course

- Understand the general functionality and technology of **blockchains**, **crypto currencies** and **public ledgers**
- Know the benefits and problem of blockchain, web3, crypto currency all technologies of that ecosystem
- Understand how **enterprise applications** can use that technologies to achieve goals
- Improve the **programming skills**

Part 1

Lecture 1

Lecture table of contents

- Understand first concepts and ideas of crypto currency
- Learn about Bitcoin in general
- Learn about Accounts, Wallets and how to use them

The History Of Bitcoin

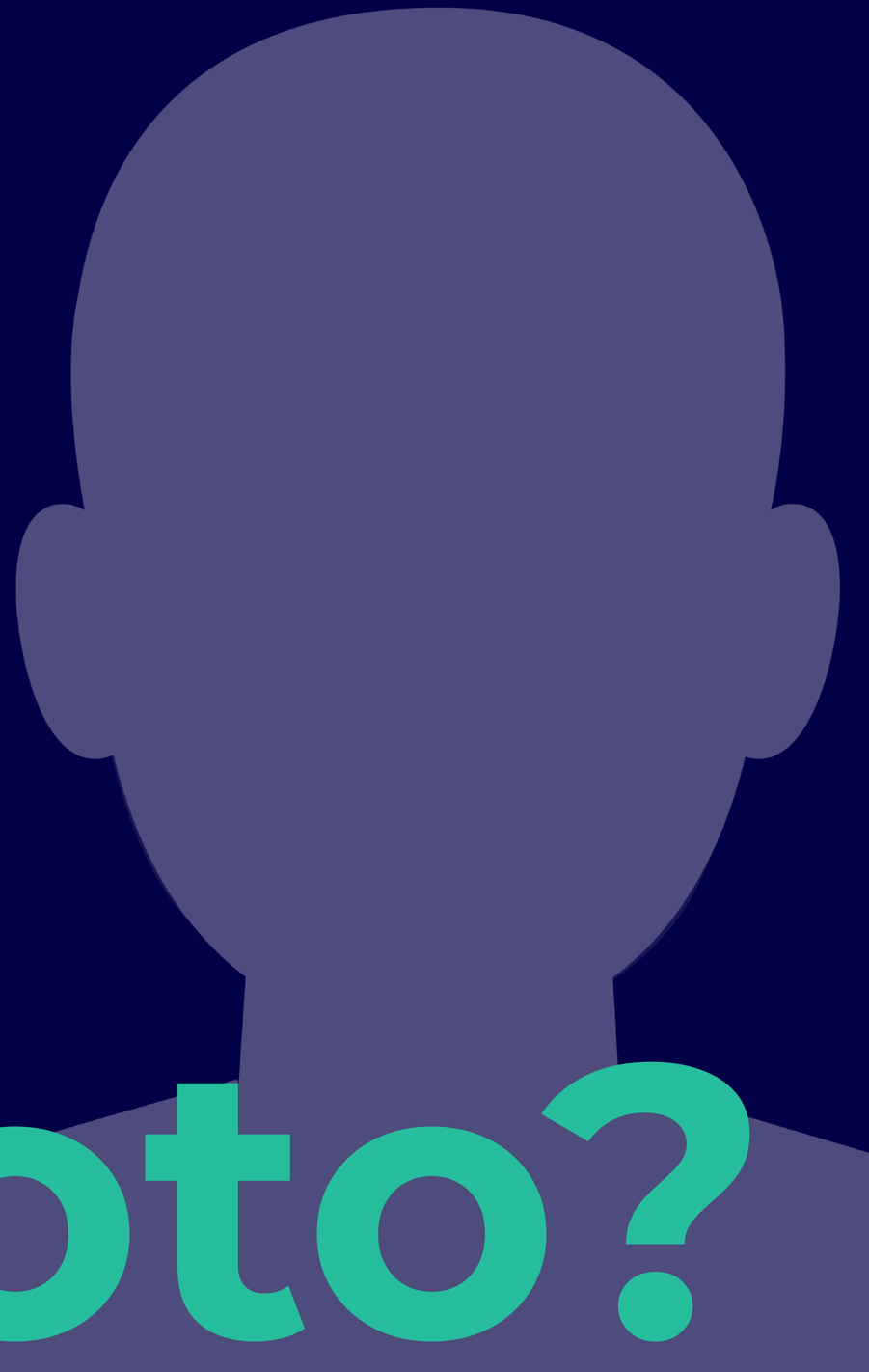
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Hands on lesson

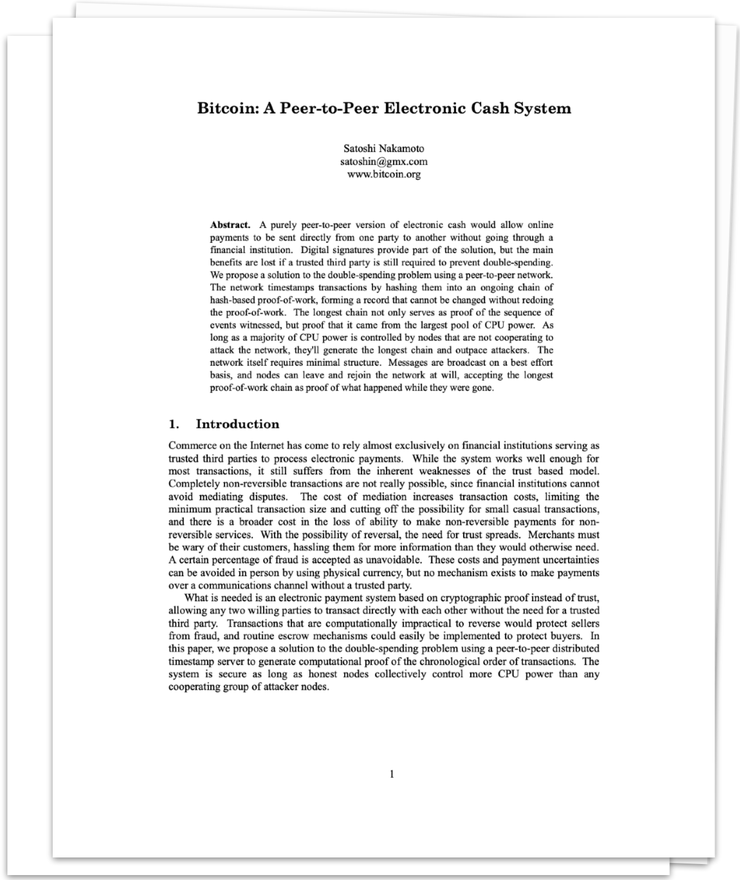
- Who owns coins of a crypto currency
- Who has a wallet for a crypto currency
- Who has no idea about about crypto currencies?

Who is Satoshi Nakamoto?



The Invention of Bitcoin

- The concept of Bitcoin has been introduced by Satoshi Nakamoto in 2008
- The paper defines the concept of the first decentralized blockchain without trusted parties /leaders



Bitcoin: A Peer-to-Peer Electronic Cash System

Satoshi Nakamoto
satoshin@gmx.com
www.bitcoin.org

Abstract. A purely peer-to-peer version of electronic cash would allow online payments to be sent directly from one party to another without going through a financial institution. Digital signatures provide part of the solution, but the main benefits are lost if a trusted third party is still required to prevent double-spending. We propose a solution to the double-spending problem using a peer-to-peer network. The network timestamps transactions by hashing them into an ongoing chain of hash-based proof-of-work, forming a record that cannot be changed without redoing the proof-of-work. The longest chain not only serves as proof of the sequence of events witnessed, but proof that it came from the largest pool of CPU power. As long as a majority of CPU power is controlled by nodes that are not cooperating to attack the network, they'll generate the longest chain and outpace attackers. The network itself requires minimal structure. Messages are broadcast on a best effort basis, and nodes can leave and rejoin the network at will, accepting the longest proof-of-work chain as proof of what happened while they were gone.

1. Introduction

Commerce on the Internet has come to rely almost exclusively on financial institutions serving as trusted third parties to process electronic payments. While the system works well enough for most transactions, it still suffers from the inherent weaknesses of the trust based model. Completely non-reversible transactions are not really possible, since financial institutions cannot avoid mediating disputes. The cost of mediation increases transaction costs, limiting the minimum practical transaction size and cutting off the possibility for small casual transactions, and there is a broader cost in the loss of ability to make non-reversible payments for non-reversible services. With the possibility of reversal, the need for trust spreads. Merchants must be wary of their customers, hassling them for more information than they would otherwise need. A certain percentage of fraud is accepted as unavoidable. These costs and payment uncertainties can be avoided in person by using physical currency, but no mechanism exists to make payments over a communications channel without a trusted party.

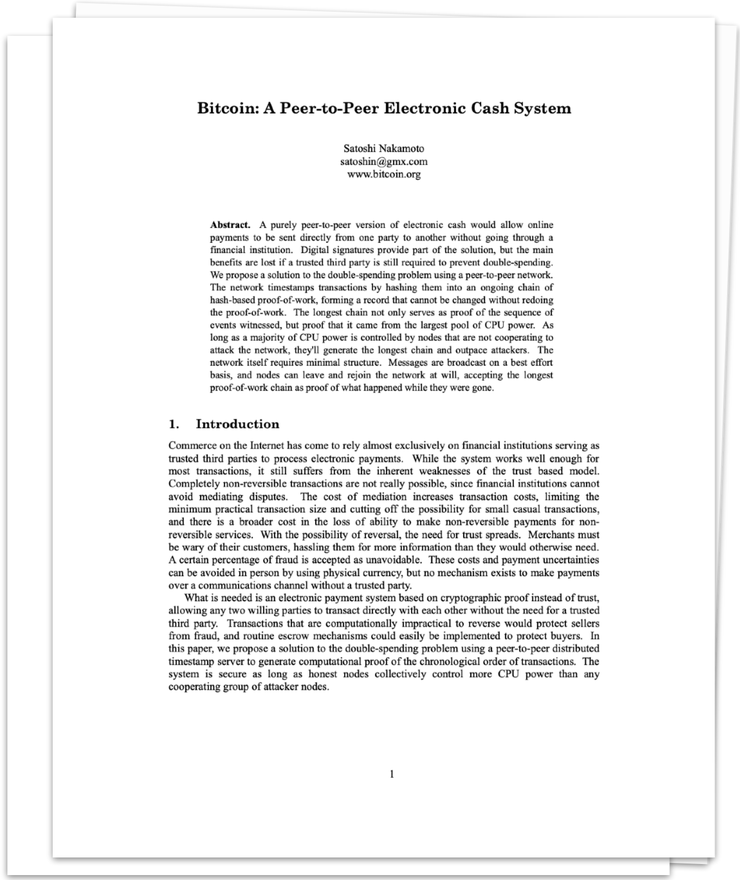
What is needed is an electronic payment system based on cryptographic proof instead of trust, allowing any two willing parties to transact directly with each other without the need for a trusted third party. Transactions that are computationally impractical to reverse would protect sellers from fraud, and routine escrow mechanisms could easily be implemented to protect buyers. In this paper, we propose a solution to the double-spending problem using a peer-to-peer distributed timestamp server to generate computational proof of the chronological order of transactions. The system is secure as long as honest nodes collectively control more CPU power than any cooperating group of attacker nodes.

1

The Invention of Bitcoin

BITCOIN.ORG

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Bitcoin: A Peer-to-Peer Electronic Cash System

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1

—What were the goals of Bitcoin?

- Create a „every day“ currency
- Have a common currency with less exchange cost
- cryptographic proof, instead of trust (against banks)
- Hard cap of coin count

21 MILLION



The Invention of Bitcoin

- Satoshi Nakamoto implemented the design in 2009
- Source can be found at GitHub
  `https://github.com/bitcoin/bitcoin`



The Inventor of Bitcoin

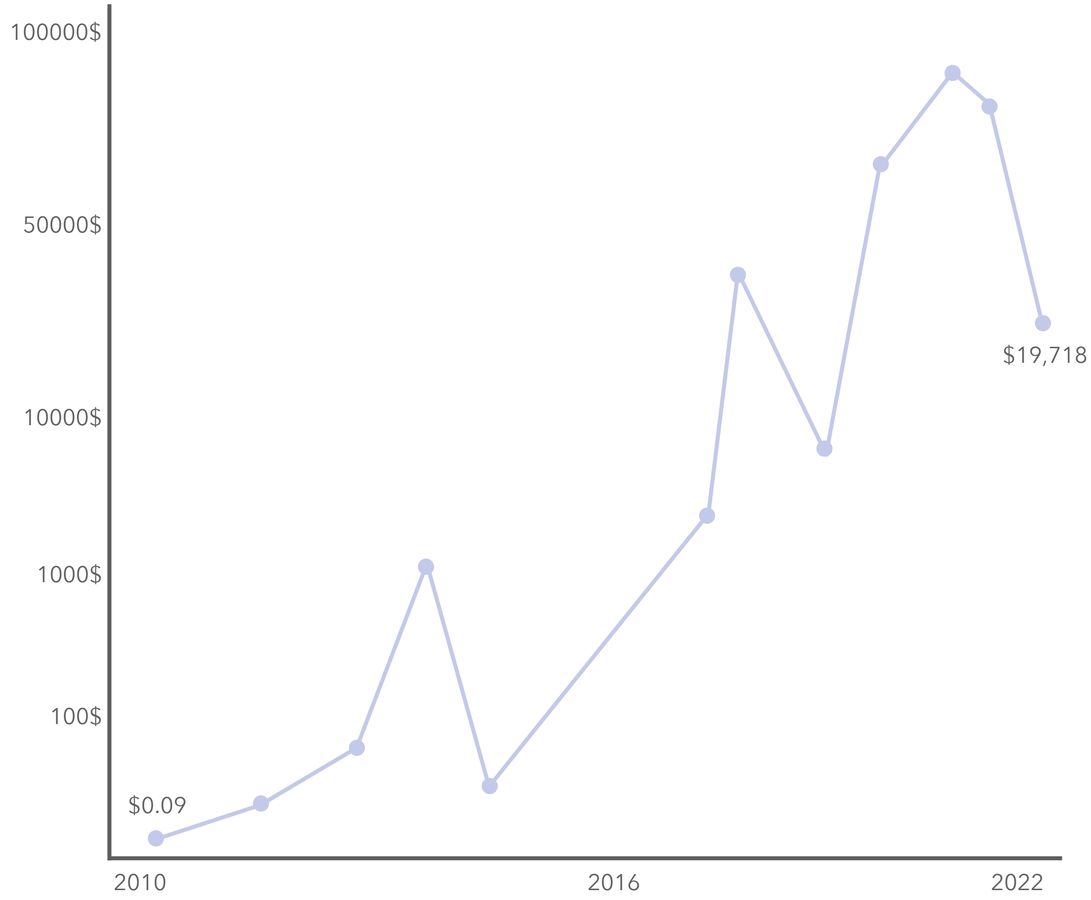
- Nakamoto became inactive in 2010
- Nobody know who Satoshi Nakamoto really is
- Over the years several people claimed to be Nakamoto



… is really rich

- Nakamoto owns between 750,000 and 1,100,000 bitcoin.
- If he is a person he is one of the richest persons in the world





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—The other side of the coin

„A 2017 report from Chainalysis, a forensics company, estimated that **between 2.78 million and 3.79 million bitcoins have been lost**.„

https://www.bankrate.com/investing/how-to-recover-lost-bitcoins-and-other-crypto/

3,000,000 ₿ ≈ 59,136,000,000 $



Bitcoin Pizza Day

- On May the 4th is Star Wars day
- On May the 22nd is Bitcoin Pizza day
- … the story started 12 years ago





—2010 at the Bitcoin forum



May 18, 2010

I'll pay 10,000 bitcoins for a couple of pizzas.. like maybe 2 large ones so I have some left over for the next day. [...] If you're interested please let me know and we can work out a deal.

Thanks,
Laszlo

https://bitcointalk.org/index.php?topic=137.0

—2010 at the Bitcoin forum



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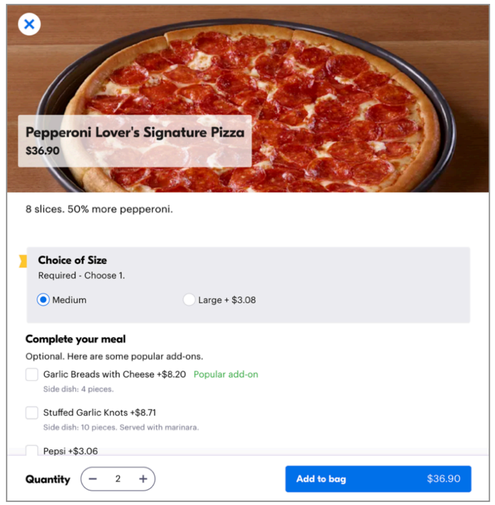
10,000... Thats quite a bit.. you could sell those on https://www.bitcoinmarket.com/ for $41USD right now.. good luck on getting your free pizza.

https://bitcointalk.org/index.php?topic=137.0

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—2010 at the Bitcoin forum



2 pizza cost about 40$ in NY Today



1 Bitcoin cost 19,000 $ today



Today the pizza would be worth 190,000,000 $

—2010 at the Bitcoin forum



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May 21, 2010

So nobody wants to buy me pizza? Is the bitcoin amount I'm offering too low?

https://bitcointalk.org/index.php?topic=137.0



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—2010 at the Bitcoin forum



May 21, 2010

I just think it would be interesting if I could say that I paid for a pizza in bitcoins 😊

https://bitcointalk.org/index.php?topic=137.0

—2010 at the Bitcoin forum

May 22, 2010

I just want to report that I successfully traded 10,000 bitcoins for pizza.



Thanks jercos!



https://bitcointalk.org/index.php?topic=137.0

—Has Bitcoin achieved the goals?

- Create a „every day“ currency
- Have a common currency with less exchange cost
- cryptographic proof, instead of trust (against banks)



—Has Bitcoin achieved the goals?

- Create a „every day“ currency ✗
- Have a common currency with less exchange cost
- cryptographic proof, instead of trust (against banks)



Digital Currencies

Coins

- **Digital or Physical Asset:** Can represent a digital cryptocurrency or a physical medium of exchange.
- **Intrinsic Value:** Coins in cryptocurrency usually have their own intrinsic value and can operate independently.

Coins

- **Medium of Exchange:** Used as a medium of exchange to facilitate transactions, both in the physical and digital realms.
- **Store of Value:** In the context of cryptocurrencies, often seen as a store of value, especially for coins like Bitcoin which is sometimes referred to as 'digital gold'.

The hype behind crypto

- Media buzz by influencers & celebrities often amplify the hype
- Many participants are driven by a fear of missing out (FOMO) on potential windfalls





The hype behind crypto

- **Financial Sovereignty:** Cryptocurrencies operate outside the traditional financial systems
- **Financial Inclusion:** Providing banking and financial services to the unbanked populations
- **Global Accessibility:** Cryptocurrencies can be accessed and traded by anyone with an internet connection



The hype behind crypto

- **Digital Gold Narrative:**
  Many view cryptocurrencies, especially Bitcoin, as **"digital gold"**, a new form of investment that may potentially offer high returns.



Stable Coins

- **Value Stability:** Designed to have a stable value, as opposed to the high volatility seen in most cryptocurrencies.
- **Pegged Value:** Often pegged to a stable asset, such as a specific amount of a fiat currency or a commodity like gold.
- **Use in Transactions:** Facilitates digital transactions by offering the speed and decentralization of cryptocurrencies without the usual volatility.

—A real usecase

- Assume you live in **South Africa** and want to **transfer money** to a person in **South Korea**



A real usecase

- **Standard Bank** is the largest African bank (by assets)
- **Shinhan Bank** is a leading bank in South Korea



—A real usecase

- **Standard Bank** is the largest African bank (by assets)
- **Shinhan Bank** is a leading bank in South Korea



- **Transaction Fee is 40 $ or even higher**
- **Transaction needs up to 6 business days**
- **Transaction can not be tracked**
- **Only provided at limited operation hours**

—A real usecase

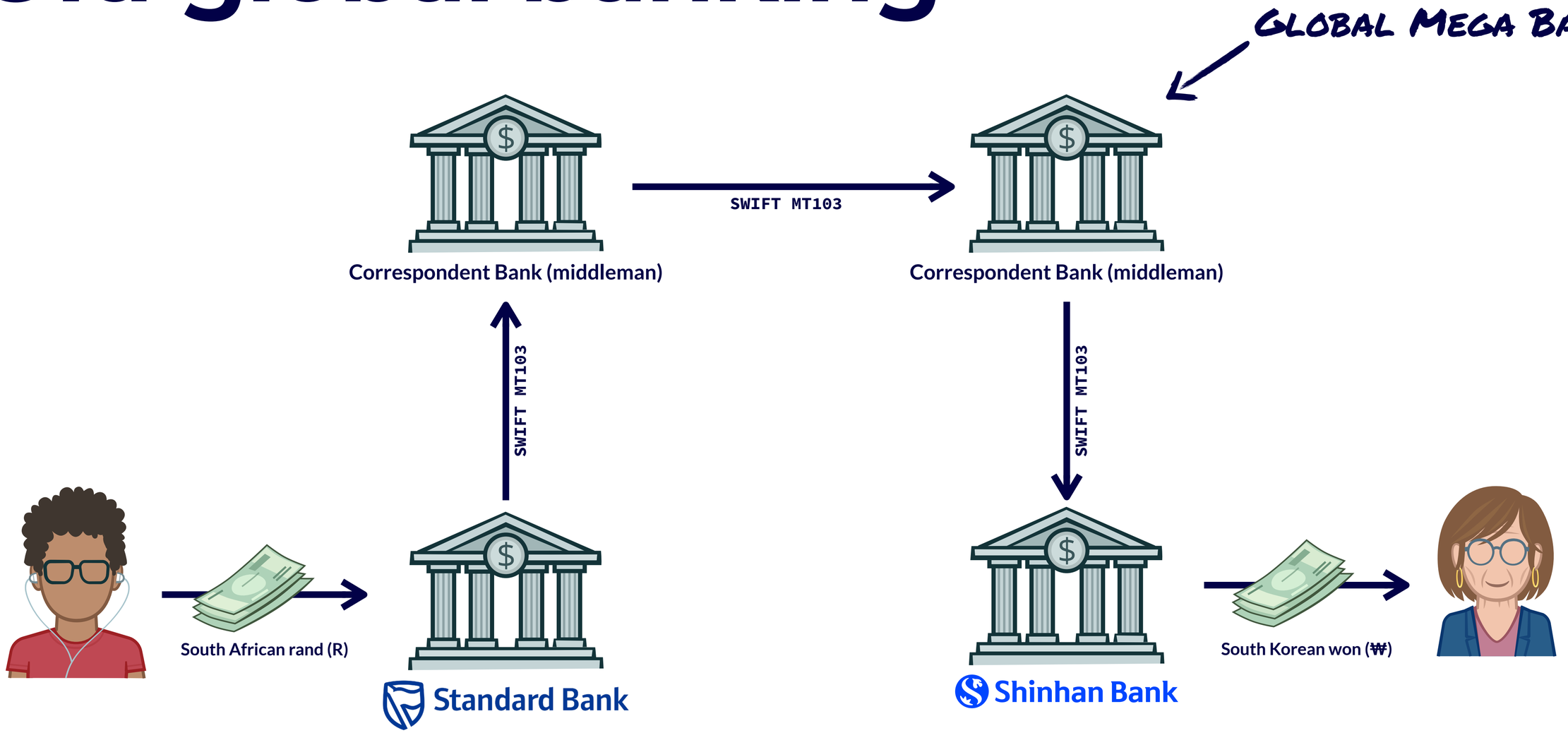
- **Standard Bank** and **Shinhan Bank** work together
- Enterprise functionalities of ledger technology is used
- Custom stable coins are used to solve the problem
- PoC based on **Hedera** released in September 2023



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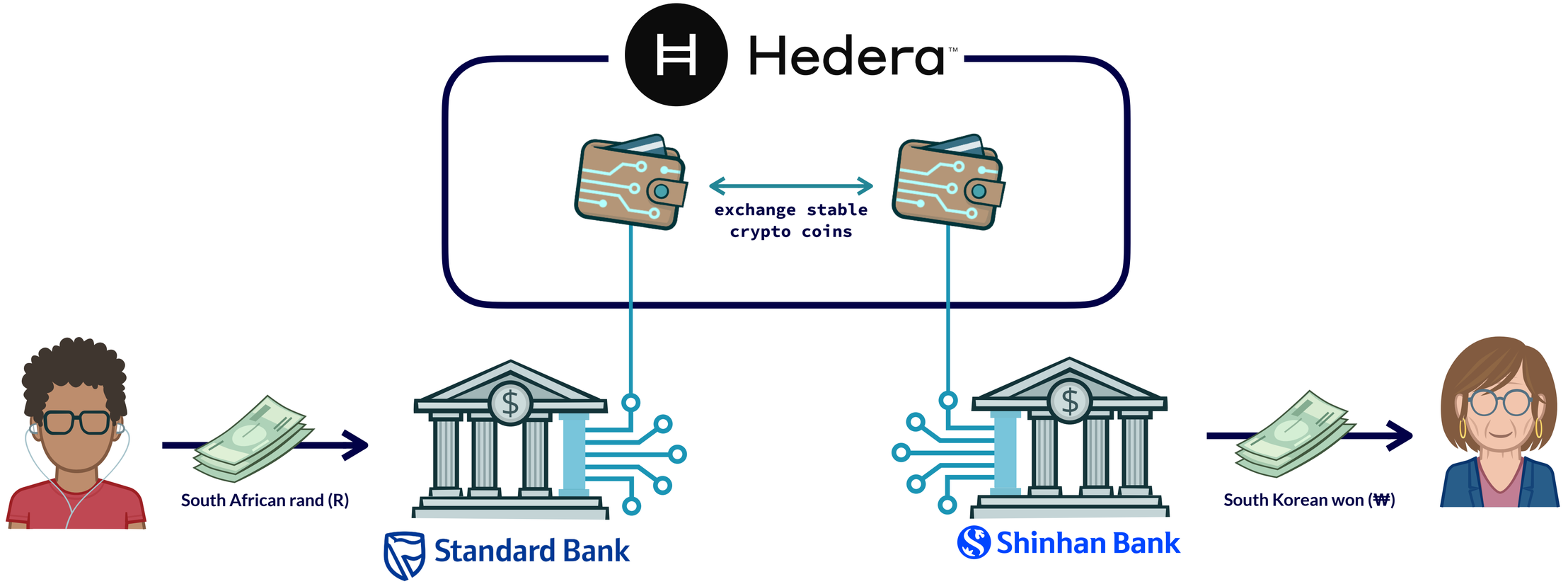
—Old global banking



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Future global banking



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A real usecase

- **Standard Bank** and **Shinhan Bank** work together
- PoC based on **Hedera**

- ~~Transaction Fee is 10 $ or even higher~~
- **Fee is less than 10 ¢**
- ~~Transaction needs up to 6 business days~~
- **Needs less than 1 min**
- ~~Transaction cannot be tracked~~
- **Real time tracking**
- ~~Only provided at limited operation hours~~
- **Provided 24/7**

—A real usecase

- **Standard Bank** and **Shinhan Bank** work together
- PoC based on **Hedera**

- **Fee is less than 10 ¢**
- **Needs less than 1 min**
- **Real time tracking**
- **Provided 24/7**

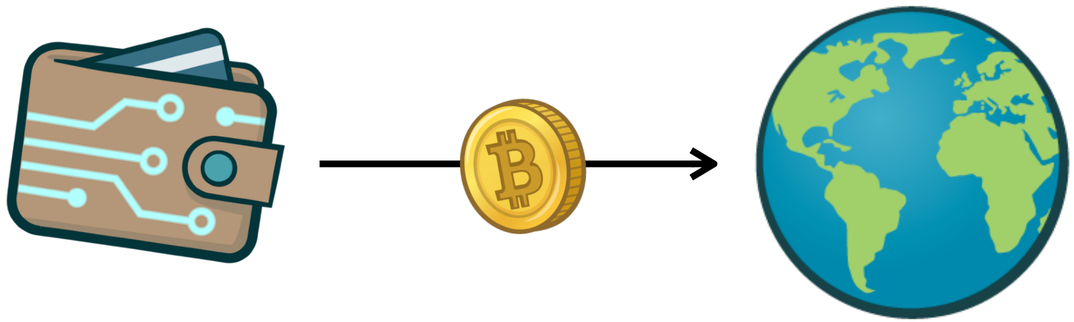


Hands on lesson

- Go to *`https://www.coinbase.com`*
- Have a look at different crypto currencies & its value over time
- Have a deeper look at the following currencies: BitCoin, Blades Of Glory, Ethereum, Lion King, USDT, Mother of Memes, HBAR, SCAM
- Let's discuss our findings

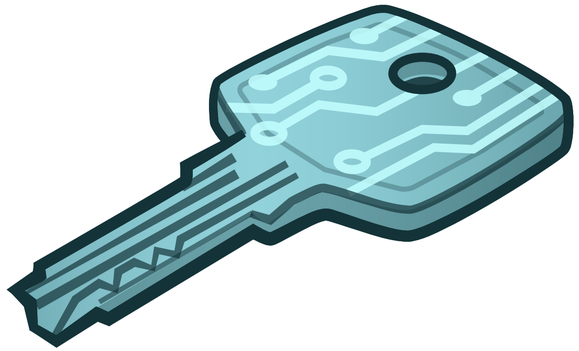
—Wallets

- Today you can not pay everywhere with bitcoin but you can trade it everywhere
- To trade coins you need a „crypto wallet“
- You will find several products on the market



—Wallets

- A wallet needs to store your public key
- A wallet needs to store your private key

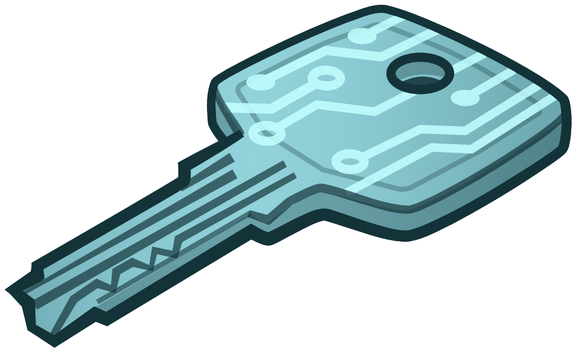


Wallets

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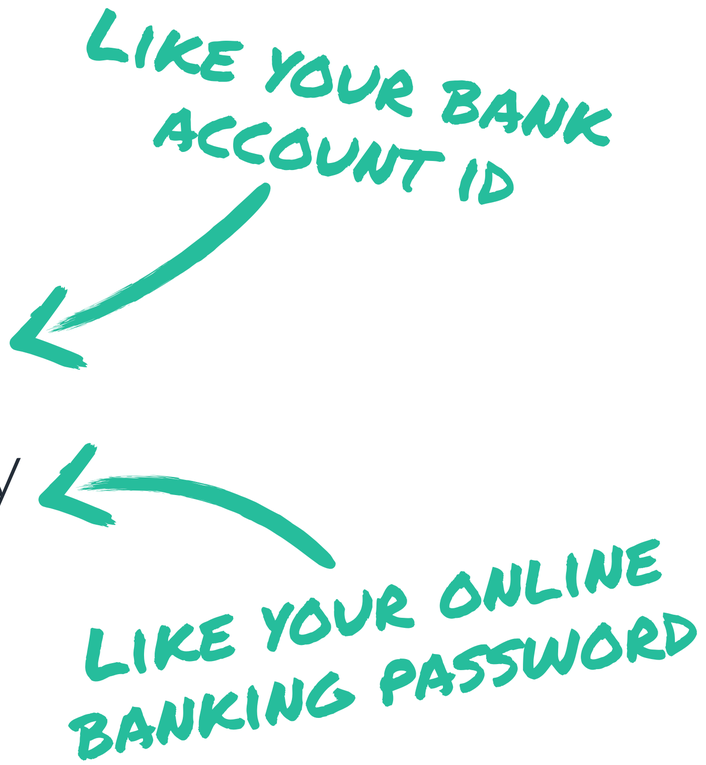
LIKE YOUR BANK ACCOUNT ID

LIKE YOUR ONLINE BANKING PASSWORD



Wallets

- A wallet needs to store your public key
- A wallet needs to store your private key



But if you miss that information no bank employee can reset your password

—Wallets

- **Digital Storage:** Wallets serve as digital storage for cryptocurrencies, allowing users to store, send, and receive digital assets.
- **Private and Public Keys:** Manages a pair of cryptographic keys: a public key, which is like an address that others see and send assets to, and a private key, which is known only to the owner and grants access to the wallet.

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Wallets

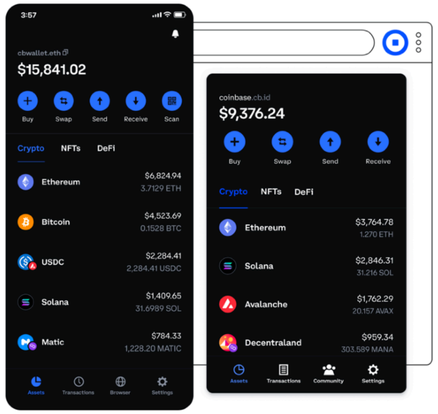
WALLET

Coinbase Wallet

Your key to the world of crypto

- Store all of your crypto and NFTs in one place
- Support for hundreds of thousands of tokens and a whole world of dapps
- Explore the decentralized web on your phone or browser
- Protect your digital assets with industry-leading security

Download Coinbase Wallet



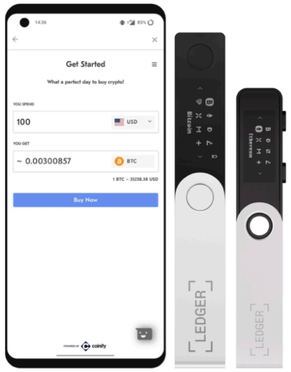
The Leading HBAR Wallet

HashPack is the gateway for Hedera dApps, DeFi and NFTs. Powered by Hedera Hashgraph.

Get HashPack







Ledger Live

ONE PLACE FOR ALL YOUR CRYPTO NEEDS

Buy

Buy Bitcoin and other cryptocurrencies – the simple, safe, smart way.

Learn about buying crypto →

Exchange

Exchange cryptocurrencies in seconds to get the most out of your assets.

Learn about exchanging crypto →

Grow

Grow your crypto assets without lifting a finger – all on one secure platform.

Learn about growing crypto →

Unlock a world of crypto possibilities

Access to all our smart services – DeFi apps, NFTs and more.

See all services →

—Wallets

- **Hardware Wallets:** Physical devices that store the user's private keys offline. Highly secure against online hacking since they are cold storage (not connected to the internet).
- **Paper Wallets:** Physical documents containing a public address for receiving cryptocurrency along with a private key for accessing the funds. Also considered a form of cold storage and secure against online hacking but vulnerable to physical damage or loss.

—Wallets

- **Desktop Wallets:** Software wallets installed on a personal computer or laptop. Only accessible from the device on which they are installed, offering a decent level of security.
- **Mobile Wallets:** Similar to Desktop Wallets. Smartphone apps that store cryptocurrency.
- **Web Wallets:** Accessed through web browsers. Convenient for quick access but potentially vulnerable to online hacking and phishing scams.

—Wallets

- **Hot Wallets:** Wallets that are connected to the internet, including web wallets and some desktop and mobile wallets. Vulnerable to online attacks but convenient for frequent transactions.
- **Cold Wallets:** Wallets not connected to the internet, including hardware and paper wallets. Provide better security against online attacks but less convenient for frequent transactions.

—Wallets

- **Security Measures:** Incorporates security features such as encryption, two-factor authentication, and backup options to protect assets and ensure secure access.
- **Multi-Signature Support:** Some wallets offer multi-signature support, requiring multiple parties to authorize a transaction, enhancing security measures.
- **Integration with Exchanges and Platforms:** Often integrated with cryptocurrency exchanges and platforms to facilitate easy and quick transactions and asset management.

Coinbase Wallet

- The Coinbase Wallet is a product of Coinbase, one of the most well-known cryptocurrency exchanges
- The wallet support multiple assets like Bitcoin (BTC) or Ether (ETH)
- Available as mobile app or chrome extension

HashPack

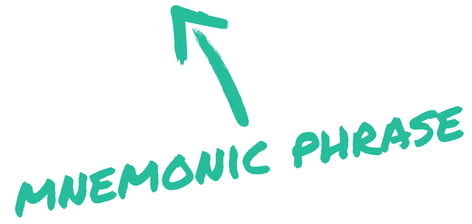
- The most used Wallet for the Hedera network
- The coin of the Hedera network is HBAR
- HashPack supports the Hedera Testnet

SHOULD BE USED TO TEST APPLICATIONS

- **Testnet** is a parallel network to the hedera network
- Users receive daily free coins on testnet

—Ledger

- Most popular and reputable **hardware wallet**
- Securely store, manage, and transact digital assets.
- Supports over 1,000 cryptocurrencies, including popular ones like Bitcoin, Ether and HBAR
- At setup a 24-word recovery phrase is defined. This phrase can restore the user's assets on any Ledger device



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Hands on lesson

- Install a wallet for the Hedera network (HashPack)
- Create an account for the Hedera Testnet
- Create a transaction that sends coins to another account

—Has Bitcoin achieved the goals?

- Create a „every day“ currency ✗
- Have a common currency with less exchange cost
- cryptographic proof, instead of trust (against banks)



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Crypto Currency

—Has ~~Bitcoin~~ achieved the goals?

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